

PRANAVI JAIN

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SUMMARY

Quantum Software Engineer with experience building modular Python frameworks for quantum experimentation, data pipelines, and circuit analysis across simulators and cloud hardware backends. Background in enterprise backend services, API development, and production backend systems. Seeking to contribute to quantum data infrastructure and software tooling in your ecosystem.

EDUCATION

University of Southern California (USC)

Master of Science in Quantum Information Science | GPA – 3.57/4

Los Angeles, CA

August 2023 – May 2025

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Electronics and Computers | CGPA – 9.03/10

Patiala, India

July 2017 – June 2021

WORK EXPERIENCE

NeoQuortex

College Park, MD

Quantum Software Engineer

June 2025 – August 2026

- Own end-to-end design and implementation of ADAPT-VQE from scratch in PennyLane across 9 fermionic and qubit-based operator pools; benchmark on H_2 (4q), LiH (6–12q), and N_2 (20q) in STO-3G basis to drive ansatz selection.
- Provide optimized ADAPT-VQE ansatzes to QUASI-NQS, a proprietary SQD-type algorithm, for sampling on IQM hardware.
- Identify the need for systematic ansatz evaluation; build a modular PennyLane framework to characterize parametrized quantum circuits across expressibility, entanglement capability, and compiled gate depth; extensible to arbitrary ansatzes.
- Co-develop a pluggable measurement module supporting standard, qubit-wise-commuting, classical shadows, and fermionic shadow strategies; implement Majorana operator algebra from first principles.
- Survey circuit knitting, IRs, and circuit synthesis; produce internal scoping documents for a planned depth reduction project.

Amdocs

Gurgaon, India

Software Developer

July 2021 – July 2023

- Led proof-of-concept engagements directly with Singtel and facilitated production releases, including Singtel RED membership.
- Delivered Java-based backend services for telecom OMS/CRM platforms to facilitate high-revenue customer deployments.
- Designed and delivered SOAP API integrations for Singtel's digital platforms within an Agile enterprise delivery cadence.
- Accelerated onboarding and training for 5+ cross-team engineers by orchestrating knowledge transfer sessions.
- Maintained code quality through peer review processes and technical documentation, improving production release readiness.

ACADEMIC PROJECTS

- **Zero-Noise Extrapolation (ZNE) for Quantum Error Mitigation** – Implemented ZNE workflow and its variants using Qiskit and Mitiq to analyze noise scaling behavior; the Double-Exponential Extrapolation variant achieved highest fidelity rates.
- **Benchmarking Quantum Computers via Grover's Algorithms** – Executed a small-scale Grover's algorithm across Aer simulator, IBM, IonQ, and IQM hardware to analyze circuit resource scaling across backends; IBM devices gave most accurate results.
- **Two-Qubit Entanglement Dynamics on IBM Quantum Systems** – Experimental study of entanglement dynamics on IBM processors by simulating Nakajima-Zwanzig and Time-Convolutionless (TCL) master equations to compare theoretical predictions against hardware noise.
- **Post-Quantum Digital Signature for IoV (IEEE ICIBT 2022)** – Evaluated applicability of CRYSTALS Dilithium (ML-DSA) for a post-quantum signature scheme for Internet of Vehicles environments; published at IEEE ICIBT 2022.

SKILLS

- **Languages:** Python (advanced), Java, C/C++ (academic), SQL, Rust (beginner)
- **Quantum:** PennyLane, Qiskit, AWS BraKet, OpenQASM, Cirq (familiar), Mitiq, NumPy, Pandas
- **Infrastructure:** Linux/Unix, Git/GitHub, GitHub Actions, REST APIs, SOAP/XML
- **Foundations:** Linear algebra, numerical optimization, variational algorithms, quantum mechanics, quantum error correction and mitigation, open quantum systems, quantum cryptography